Date: Day:

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Top 5 Testing Types In IT Companies

1- Unit Testing

Description : Testing individual components or functions of code in isolation to verify that each unit works correctly

AI Agents / AI tools Devising Test

- Github Copilot / Copilot Workspaces
  Generates unit test directly in IDE
- Diffblue Cover :
  AI agent for Java to automatically writes complete, executable unit test
- Codium AI (qodo)
  Analyze code logic and edge cases to auto-generate unit test suites.

2- Integration Testing

Description : Testing interactions and communication between integrated modules, databases, micro-services or external API endpoints

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AI Agent / AI Tools Devising Tests

- Postman Postbot
  AI assistant inside Postman that auto-generates API integration test scripts
- Katalon AI agent
  Devises multi-layer API and system integration test flows
- Testim.io
  Uses AI to discover service relationships, record API endpoints.

3- End-to-End / functional UI Testing

Description : Simulating real user journeys across the application user interface to verify that user workflows functions as expected from start to finish

AI Agent / AI Tools Devising Tests

- Testrigor
  Generates E2E test scripts from plain English
- Functionalize
  Uses deep learning AI agents to create self healing E2E test scripts.
- Mabl
  Intelligent test automation platform using

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AI agents to generate user journeys

4- **Performance And Load Testing**

Description

Assessing system responsiveness, stability, scalability and resource consumption under specified workload conditions.

AI Agents / AI Tools Devising Test

- k6 AI / Grafana Cloud AI
  Generates k6 load testing scripts and workload scenarios from user traffic

- Dynatrace Davis AI
  AI engine that devises load test baselines, analyzes performance regressions.

- OctoPerf AI
  Generates virtual user scenarios and traffic load profiles automatically.

5- **Security Testing (DAST / SAST AND Penetration Testing)**

Description : Identifying vulnerabilities security flaws, authorization bugs and threats in application code and running enviroments.

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AI Agents / AI Tools Devising Test

- Snyk Code AI / Deepcode
  AI agent that scans codebase pattern, predicts security risks and generates targetted security regression test cases.

- Pentest GPT
  LLM-powered pentesting AI agent built on top of GPT models to devise attack vectors and vulnerability test prompts.

- Veracode Fix / AI Security Agent
  Automatically drafts security fixes and generates verification test suites for common vulnerabilities